data availability
- open data
- data on request
- closed data
- preprint
- reanalysis on open data

decoding mode
- online
- offline

electrode type
- ECoG
- Utah array
- thin-film threads
- microwire array

speech ability
- participants who can speak naturally
- participants who have vocal paralysis

task instruction
- inner speech
- overt/mouthing/intended

decoded output
- brain-to-voice
- brain-to-text

Speech

Jude et al., 2026b ◆/◇
intended speech in LIS ⬠, WER 66.5%, vocab 125k (n=1, T17△)

Feghhi et al., 2026 ◇
LightBeam ⬠, WER 5.01%/2.36%, vocab 100k (n=2, T12△/T15△)

Kunz et al., 2025 ◆/◇
inner speech ⬠, WER 23.7%/51.3%/46.6%, vocab 125k (n=3, T12△/T15△/T16△)

Brosler et al., 2026 ◇
speech+gesture avatar ⬠, WER 30.0%, vocab 5-10 (n=3, B1r△/B3△/B6△)

Card et al., 2024 ◇
rapid calibration ⬠, WER 4.9%, vocab 125k (n=1, T15△)

Boccato et al., 2026 ◆/◇
cross-subject CTC ⬠, WER 10.3%/5.9%, vocab 125k (n=4, T12△/T15△/T16△/T17△)

Wairagkar et al., 2026 ◇
Brain2voice 2.0 ⬟, listener WER 5.24%, open vocab (n=1, T15△)

Willett et al., 2023 ◇
large-vocab decoding ⬠, 62 wpm, WER 23.9%, vocab 125k (n=1, T12△)

Littlejohn et al., 2025 ◇
streaming voice ⬟, listener WER 58.8%, vocab 1,024 (n=1, B3△)

Y. Zhang et al., 2025 ◆/◇
Brain-to-Text (BIT) ⬠, WER 5.10%/1.76% (E2E 10.22%/7.76%), vocab 125k (n=2, T12△/T15△)

Metzger et al., 2022 ◇
NATO code-word spelling ⬠, WER 10.5%, vocab 1,152 (n=1, B1△)

Wandelt et al., 2024 ◆/◇
internal speech ⬠, WER 21%/77%, vocab 8 (n=2, P1▲/P2▲)

Levin et al., 2026 ◇
cross-brain transfer ⬠, PER 23-40% (WER n/r), vocab 125k (n=3, T12△/T15△/T16△)

Card et al., 2026 ◇
3,800 h home ⬠, WER 0.8-3.5%, vocab 125k (n=1, T15△)

n = 16 unique participants

Moses et al., 2021 ◇
real-time sentence decoding ⬠, 15.2 wpm, WER 25.6%, vocab 50 (n=1, B1△)

Metzger et al., 2023 ◇
78 wpm + avatar ⬟/⬠, WER 25.5%, vocab 1,024 (n=1, B3△)

Wairagkar et al., 2025 ◇
voice synthesis ⬟, listener WER 43.8%, open vocab (n=1, T15△)

Yoon et al., 2026 ◇
real-time ensembles ⬠, WER 33.7→26.0%, vocab 125k (n=1, T12△)

Paradromics, 2026.09.14 ◇
real-time speech ⬠, WER n/r, vocab n/r (n=1, C1△)

Angrick et al., 2024 ◇
speech synthesis ⬟, listener WER 20%, vocab 6 (n=1, CC01△)

Li et al., 2025 ◇
DConD-LIFT ⬠, WER 5.77% (1-model 8.06%), vocab 125k (n=1, T12△)

Fogg et al., 2026 ◇
multi-user ⬠, WER 3.9-8.0%, vocab 125k (n=6, T12△/T15△/T16△/T17△/T21△/T22△)

Silva et al., 2024b ◇
bilingual ⬠, WER 25.0%, vocab 178 (n=1, B1△)

Feghhi et al., 2025 ◇
time-masked Transformer ⬠, WER 7.98% (ens. 5.68%), vocab 125k (n=1, T12△)

Z. Zhang et al., 2026 ◇
ALIGN cross-day ⬠, unseen-day WER 32.2%/45.5%, vocab 125k (n=2, T12△/T15△)

Feng et al., 2024 ◇
E2E RNN-LLM ⬠, WER 26.3% (B2T 24.69%), vocab 125k (n=1, T12△)

Nason-Tomaszewski et al., 2026 ◇
Q&A 27.7 wpm ⬠, WER 35.2%, vocab 125k (n=1, T16△)

Pouratian et al., 2026 ◇
fully implanted speech BCI ⬠, 75 wpm, WER 5.8%, vocab 125k (n=3, V1△/V2△/V3△)

Typing

Neuralink, 2026.01.28 ◇
ten-finger typing ⬠, 40 wpm, WER n/r, vocab n/r (n=1, P7▲)

Jude et al., 2026a ◇
touch typing neuroprosthesis ⬠, 110 cpm, WER 12.5%/3.4%, vocab 125k (n=2, T17△/T18▲)

n = 3 unique participants

Handwriting

Karpowicz et al., 2024 ◇
FALCON H2 benchmark ⬠, WER 11.0% (held-in 2.0%), vocab 130k (n=1, T5▲)

Levin et al., 2026 ◇
cross-brain transfer ⬠; CER 2% (WER n/r), open vocab (n=3, T5▲/T12△/T18▲)

Willett et al., 2021 ◇
brain-to-text ⬠, 90 cpm, WER 25.1%, open vocab (n=1, T5▲)

Fan et al., 2023 ◇
403-day plug-and-play ⬠, WER 4.0%, vocab 130k (n=1, T5▲)

Xu et al., 2025 ◇
video-cued trajectory decoding ⬠, CER 8.9% (WER n/r), vocab 1,000 (n=1, Z1▲)

Yang et al., 2026 ◇
online Chinese character decoding ⬠, 8.55 cpm, CER 5.6%, vocab 295 (n=1, Z1▲)

n = 4 unique participants

2021 2022 2023 2024 2025 2026 2027

20 unique participants in total; 1 used a system at home (T15)